

PART 1: INTRODUCTION TO SIEM

Research and explain:

- 1. What is SIEM?**
- 2. Why do organizations use SIEM?**
- 3. Difference between SIEM and Log Management.**
- 4. Popular SIEM Tools:**
 - **Splunk**
 - **IBM QRadar**
 - **Microsoft Sentinel**
 - **ELK Stack**

1. What is SIEM?

Security Information and Event Management (SIEM) is a cybersecurity solution that combines Security Information Management (SIM) and Security Event Management (SEM) to provide centralized collection, storage, monitoring, analysis, and management of security-related data. It gathers logs and events from various sources such as servers, workstations, network devices, firewalls, applications, databases, and cloud services into a single platform.

A SIEM system continuously analyzes this data to identify suspicious activities, detect security threats, generate real-time alerts, and assist security teams in responding to incidents. By correlating events from multiple sources, SIEM provides a comprehensive view of an organization's security posture and helps detect attacks that might otherwise go unnoticed.

Modern SIEM solutions also incorporate features such as threat intelligence integration, automated incident response, user behavior analytics, compliance reporting, and customizable dashboards, making them an essential component of a Security Operations Center (SOC).

Key Functions of SIEM

- Collects logs from multiple devices and applications.
- Stores and indexes log data for future analysis.
- Correlates events from different sources.
- Detects suspicious activities and security incidents.
- Generates real-time alerts.
- Provides dashboards and visual reports.
- Supports incident investigation and digital forensics.
- Helps organizations meet regulatory compliance requirements.

2. Why Do Organizations Use SIEM?

Organizations generate millions of log entries every day from servers, operating systems, firewalls, antivirus software, web applications, cloud platforms, and network devices. Manually monitoring these logs is time-consuming, inefficient, and prone to human error. SIEM addresses this challenge by automating the collection, analysis, and correlation of security events from across the organization's infrastructure.

SIEM enables security analysts to monitor systems continuously, detect cyber threats quickly, investigate incidents efficiently, and respond before attackers can cause significant damage. It also helps organizations comply with industry standards and regulatory requirements by maintaining centralized log records and generating compliance reports.

Benefits of Using SIEM

1. Centralized Log Collection

SIEM collects logs from multiple sources and stores them in a centralized repository, making it easier for analysts to monitor and investigate events.

2. Real-Time Threat Detection

It continuously monitors incoming events and identifies suspicious activities such as malware infections, brute-force attacks, unauthorized access attempts, and insider threats.

3. Event Correlation

SIEM correlates related events occurring across different systems to identify complex attack patterns that may not be visible when examining individual logs.

4. Faster Incident Response

By generating immediate alerts, SIEM enables security teams to investigate and respond to incidents quickly, reducing potential damage.

5. Compliance Reporting

Many regulations, including ISO 27001, PCI DSS, HIPAA, and GDPR, require organizations to maintain security logs. SIEM simplifies compliance by storing logs securely and generating automated reports.

6. Digital Forensics

Historical logs stored by SIEM assist investigators in reconstructing security incidents, identifying attack timelines, and collecting evidence during forensic investigations.

7. Security Monitoring

SIEM provides continuous monitoring of the organization's IT infrastructure, allowing analysts to identify abnormal behavior before it becomes a major security incident.

8. Improved Visibility

Security teams gain a comprehensive view of network activity, user behavior, authentication events, and application performance through interactive dashboards and reports.

3. Difference Between SIEM and Log Management

Although both SIEM and Log Management deal with collecting logs, their purpose and capabilities differ significantly.

Feature	SIEM	Log Management
Purpose	Detects, analyzes, and responds to security threats	Collects, stores, and organizes log data
Log Collection	Yes	Yes
Log Storage	Yes	Yes
Real-Time Monitoring	Yes	Limited or No
Event Correlation	Yes	No
Threat Detection	Yes	No
Alert Generation	Yes	Usually No
Security Analytics	Advanced analytics and correlation	Basic searching and filtering
Dashboard	Interactive security dashboards	Basic log visualization
Incident Response	Supports investigation and response	Limited support
Compliance Reporting	Yes	Basic reporting
Primary Users	Security Analysts, SOC Teams	System Administrators, IT Teams

4. Popular SIEM Tools

4.1 Splunk Enterprise

Splunk Enterprise is one of the most popular commercial SIEM platforms used by organizations worldwide. It collects and indexes machine-generated data from servers, applications, operating systems, network devices, and cloud services. Using Splunk Processing Language (SPL), analysts can search, analyze, visualize, and investigate security events efficiently.

Features

- Centralized log collection and indexing
- Powerful SPL search language
- Real-time monitoring and alerting
- Interactive dashboards and reports
- Event correlation
- Threat detection and investigation
- Machine learning and predictive analytics
- Scalable architecture

Advantages

- Fast data searching
- Easy dashboard creation
- Supports multiple data sources
- Extensive integration with third-party security tools
- Strong community support and documentation

Limitations

- Commercial licensing can be expensive.
- Requires significant system resources for large deployments.
- Advanced SPL queries require training.

4.2 IBM QRadar

IBM QRadar is an enterprise-grade SIEM platform that provides security monitoring, event correlation, network traffic analysis, and threat intelligence. It is widely used by large organizations to detect advanced cyber threats and automate incident investigations.

Features

- Automatic log collection
- Event correlation
- Risk scoring
- Network flow monitoring
- Threat intelligence integration
- Vulnerability assessment
- Compliance reporting

Advantages

- Excellent threat detection capabilities
- Built-in network analysis
- Strong correlation engine
- Suitable for large enterprise environments

Limitations

- Higher implementation and licensing costs.
- Requires experienced administrators for deployment and maintenance.

4.3 Microsoft Sentinel

Microsoft Sentinel is a cloud-native SIEM and Security Orchestration, Automation, and Response (SOAR) solution hosted on Microsoft Azure. It provides intelligent security analytics, automated investigation, and threat hunting using artificial intelligence.

Features

- Cloud-native architecture
- AI-driven threat detection
- Built-in automation with playbooks
- Integration with Microsoft Defender and Azure services
- Threat hunting
- Compliance reporting

- Scalable cloud infrastructure

Advantages

- No on-premises hardware required
- Easy integration with Microsoft ecosystem
- Automatic updates and scalability
- Supports automated response workflows

Limitations

- Best suited for organizations using Microsoft Azure.
- Costs may increase depending on the volume of ingested data.

4.4 ELK Stack (Elasticsearch, Logstash, Kibana)

ELK Stack is an open-source platform commonly used for centralized log management and security monitoring. It consists of Elasticsearch for storing and searching data, Logstash for collecting and processing logs, and Kibana for data visualization and dashboards.

Components

- Elasticsearch: Stores and indexes log data for fast searching.
- Logstash: Collects, processes, and forwards log data from multiple sources.
- Kibana: Provides dashboards, charts, and visualization tools for analyzing log data.

Features

- Open-source and customizable
- Centralized log collection
- Powerful search functionality
- Interactive dashboards
- Real-time visualization
- Scalable architecture

Advantages

- Free and open source
- Highly flexible and customizable
- Strong community support
- Suitable for organizations of different sizes

Limitations

- Requires manual configuration and maintenance.
- Advanced security features may require additional components or commercial offerings.
- Steeper learning curve for beginners.

Conclusion

Security Information and Event Management (SIEM) has become an essential component of modern cybersecurity operations. It enables organizations to collect and analyze security logs from multiple sources, detect threats in real time, investigate incidents, and respond quickly to cyberattacks. Compared to traditional log management, SIEM offers advanced capabilities such as event correlation, threat detection, automated alerting, and security analytics. Popular SIEM platforms such as Splunk Enterprise, IBM QRadar, Microsoft Sentinel, and the ELK Stack provide powerful tools for monitoring and protecting organizational infrastructure. By improving visibility into security events and supporting compliance requirements, SIEM helps organizations strengthen their overall security posture and reduce the impact of cyber threats.

PART 2: SPLUNK INSTALLATION

1. **Install Splunk Enterprise (Free Trial).**
2. **Launch Splunk.**
3. **Explore the following sections:**
 - **Search & Reporting**
 - **Dashboards**
 - **Alerts**
 - **Reports**

Take screenshots.

Introduction

Splunk Enterprise is a powerful Security Information and Event Management (SIEM) platform used to collect, index, search, analyze, and visualize machine-generated data from various sources such as servers, operating systems, applications, firewalls, databases, and network devices. It enables security analysts to monitor security events in real time, investigate incidents, create dashboards, and generate alerts.

In this practical task, Splunk Enterprise was installed on a Windows system, launched successfully, and its major features—including Search & Reporting, Dashboards, Alerts, and Reports—were explored.

2.1 System Requirements

Before installing Splunk Enterprise, ensure that the system meets the following minimum requirements:

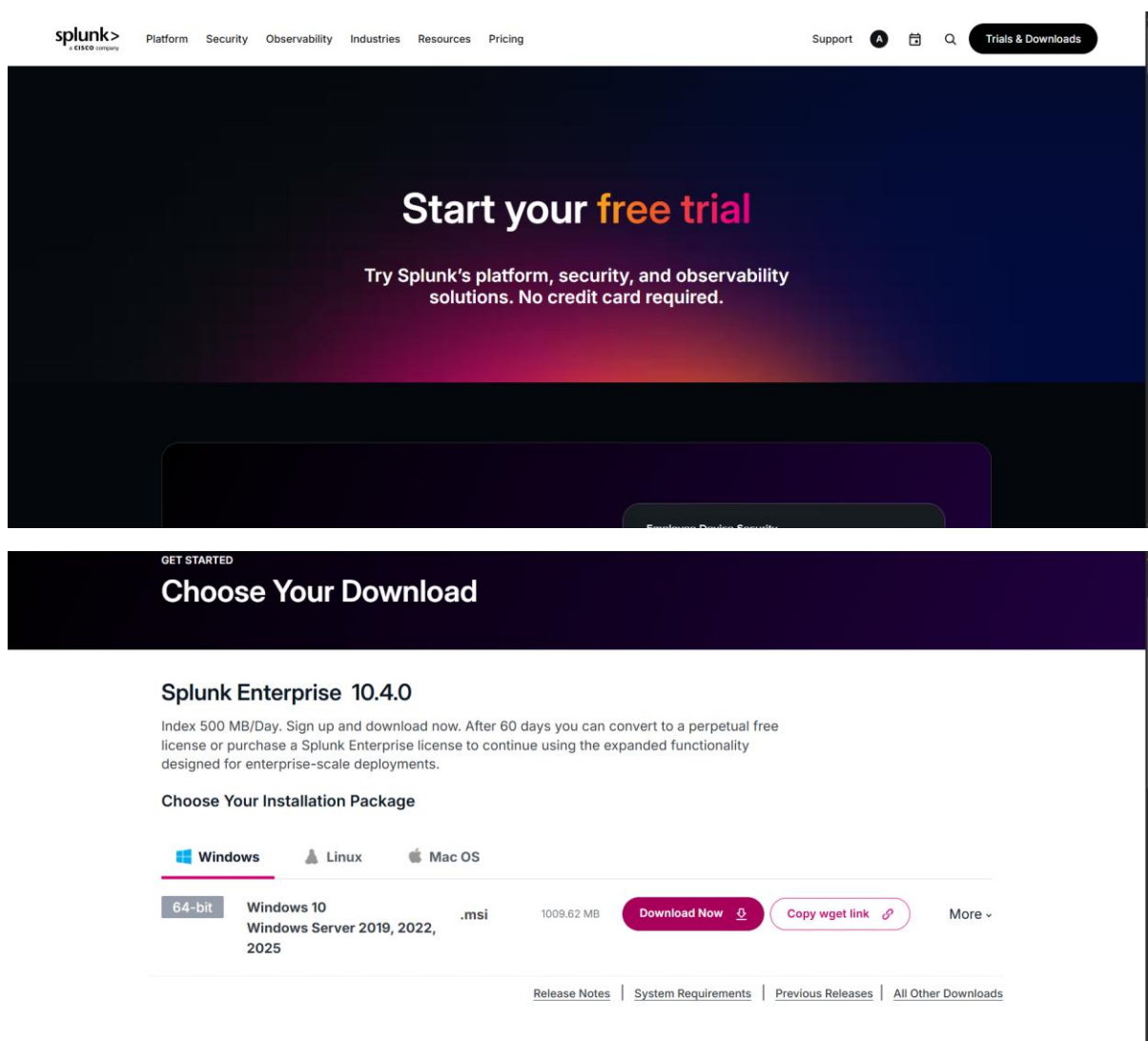
Component	Requirement
Operating System	Windows 10/11 (64-bit) or Windows Server
Processor	Intel Core i3 or higher
RAM	Minimum 4 GB (8 GB recommended)
Storage	At least 10 GB free disk space
Browser	Google Chrome, Microsoft Edge, or Mozilla Firefox
Internet	Required for downloading Splunk

2.2 Downloading Splunk Enterprise

Steps:

1. Open a web browser.
2. Visit the official Splunk website.
3. Navigate to the **Splunk Enterprise Free Trial** download page.
4. Create a Splunk account or sign in if an account already exists.
5. Select the Windows (64-bit) installer.
6. Download the installer (.msi file).

Screenshot

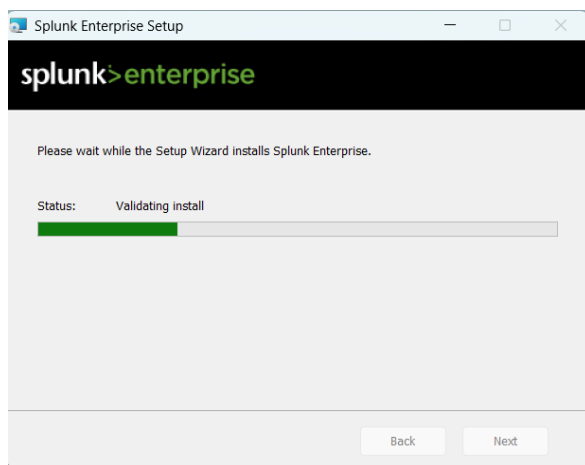


2.3 Installing Splunk Enterprise

Installation Steps

1. Double-click the downloaded **Splunk Enterprise (.msi)** installer.
2. Accept the License Agreement.
3. Select the installation directory (default location is recommended).
4. Create an administrator username and password.
5. Click **Install**.
6. Wait for the installation process to complete.
7. Click **Finish** after successful installation.

Screenshot :



2.4 Launching Splunk Enterprise

After installation:

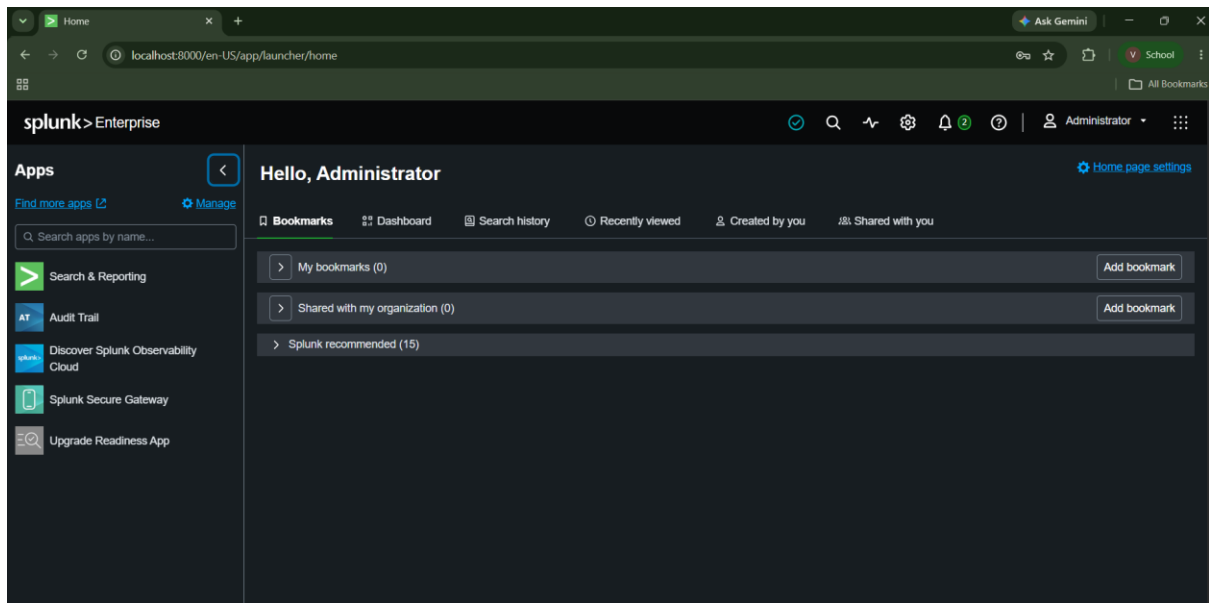
1. Open **Splunk Enterprise** from the Start Menu.
2. Splunk starts its services automatically.
3. A browser window opens automatically.
4. If it does not open automatically, visit:

<http://localhost:8000>

5. Enter the administrator username and password created during installation.
6. Click **Sign In**.

After successful login, the Splunk Home Dashboard is displayed.

Screenshot :



2.5 Exploring the Splunk Interface

After logging in, the main features of Splunk were explored.

A. Search & Reporting

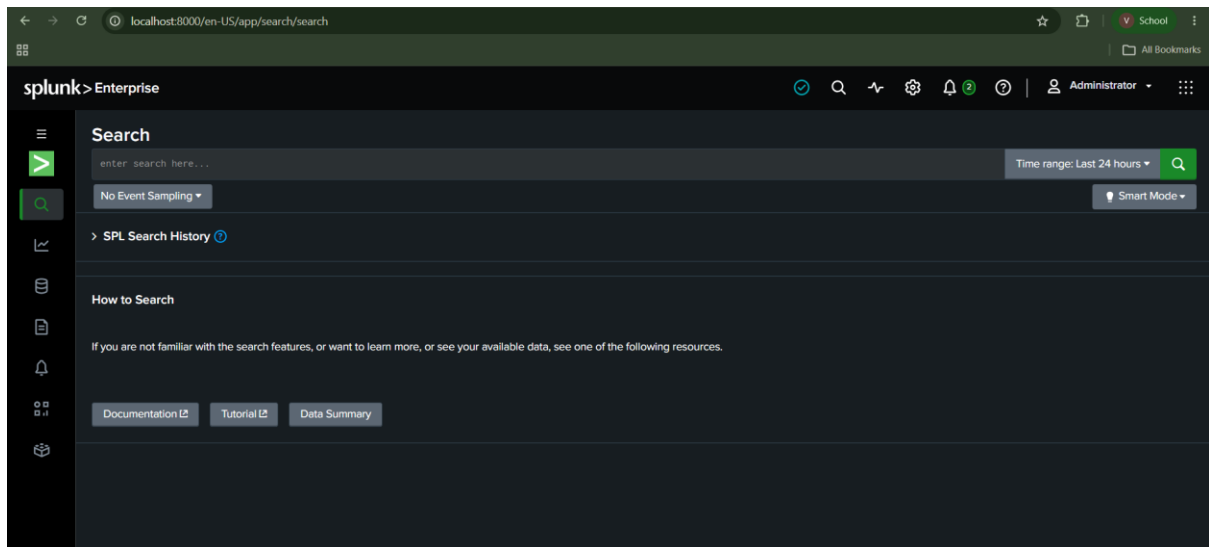
The **Search & Reporting** application is the primary workspace where users search, analyze, and investigate machine data. Security analysts use the Splunk Processing Language (SPL) to filter logs, identify suspicious events, generate statistics, and create visualizations.

Key Features

- Execute SPL queries
- Search indexed logs
- Filter events
- Generate charts and graphs
- Perform statistical analysis
- Save searches
- Create reports

Purpose: To analyze log data and investigate security events efficiently.

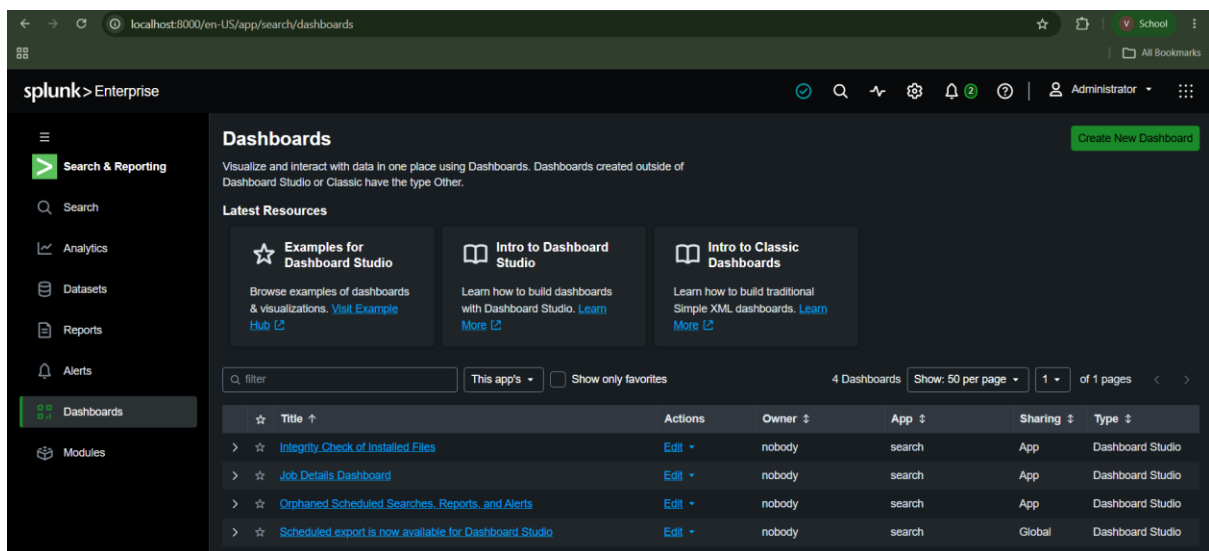
Screenshot :



B. Dashboards

Dashboards provide a graphical representation of security data using charts, tables, and visualizations. They help analysts monitor system activity and identify trends quickly.

Screenshot :



C. Alerts

Alerts allow Splunk to notify security analysts whenever predefined conditions are met. Alerts help detect suspicious activities such as repeated failed login attempts, unauthorized access, or unusual network traffic.

Types of Alerts

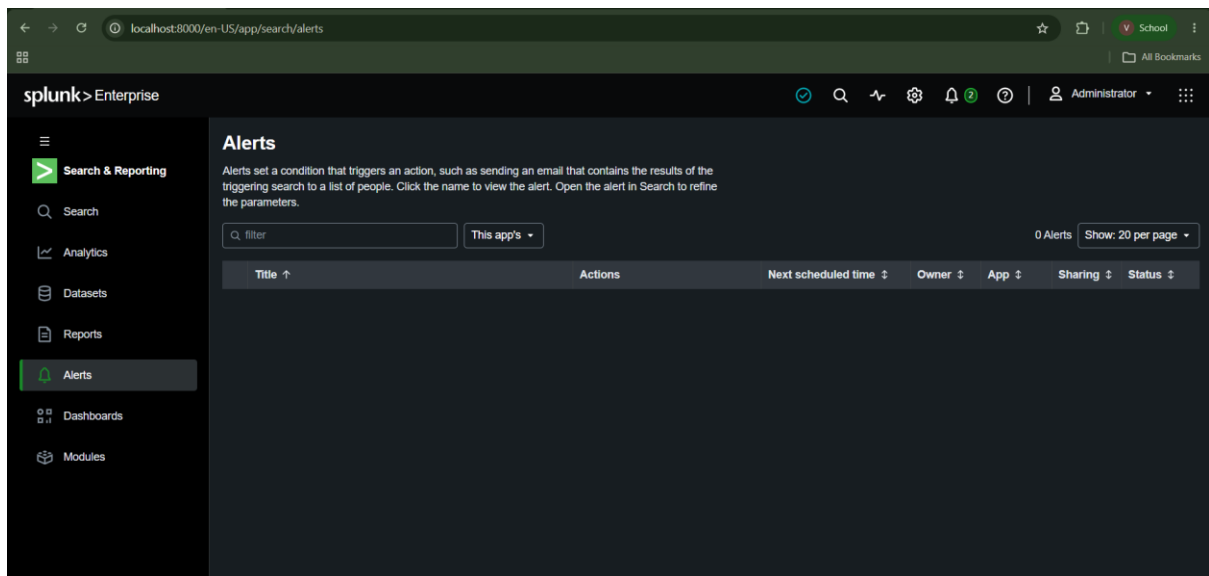
- Scheduled Alerts
- Real-Time Alerts

Alert Actions

- Send email notifications
- Trigger webhooks
- Execute scripts
- Create tickets
- Generate notifications

Purpose: To automatically detect and respond to potential security incidents.

Screenshot :



D. Reports

Reports provide summarized information based on saved searches and statistical analysis. They can be generated manually or scheduled to run automatically at specific intervals.

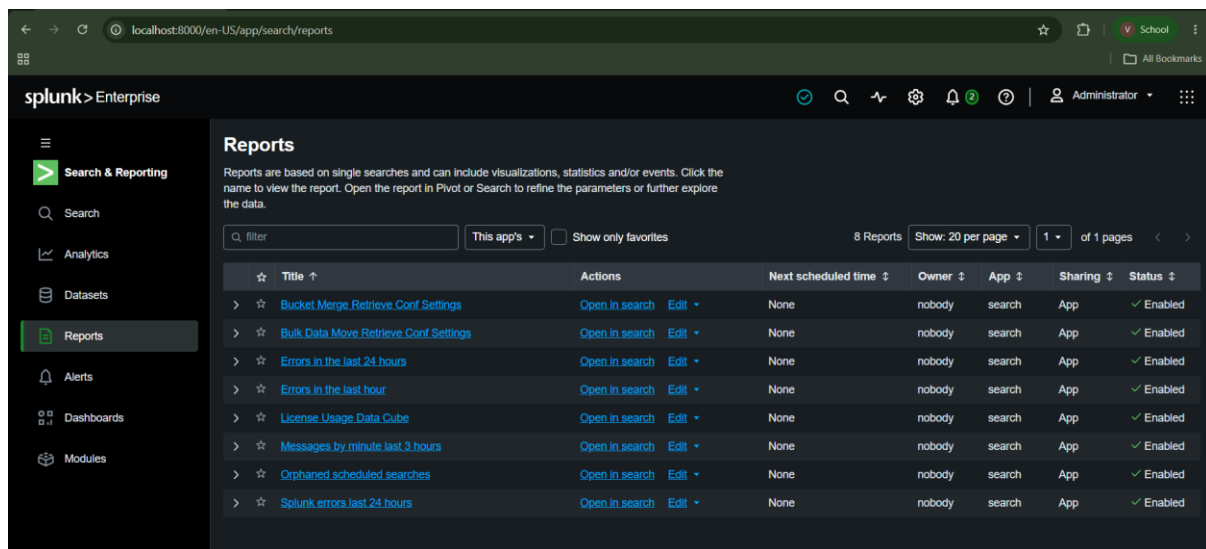
Types of Reports

- Daily reports
- Weekly reports
- Monthly reports
- Security reports
- Compliance reports
- Incident reports

Benefits

- Simplifies data analysis
- Supports auditing
- Helps management understand security trends
- Assists in compliance reporting

Screenshot :



Conclusion

Splunk Enterprise was successfully installed and launched on the Windows system. After logging in, the core features of the platform—including **Search & Reporting**, **Dashboards**, **Alerts**, and **Reports**—were explored. These components provide powerful capabilities for collecting, analyzing, visualizing, and monitoring machine-generated data.

PART 3: DATA INGESTION

Upload any sample log file into Splunk.

Examples:

- **Web Server Logs**
- **Windows Logs**
- **Application Logs**

Tasks:

1. **Import the log file.**
2. **Verify data ingestion.**
3. **Count total events.**

Take screenshots.

Introduction

Data ingestion is the process of collecting and importing log data from different sources into Splunk for storage, indexing, and analysis. Splunk supports log files from operating systems, web servers, applications, firewalls, and network devices. Once the data is indexed, it becomes searchable using the Splunk Processing Language (SPL), allowing security analysts to investigate events, detect threats, and generate reports.

Objective

The objectives of this practical are:

- Import a sample log file into Splunk.
- Verify that the data has been successfully indexed.
- Count the total number of events.
- Prepare the data for further security analysis.

Steps to Upload a Log File

Step 1: Open Splunk Enterprise

Launch Splunk Enterprise and log in using the administrator credentials.

Step 2: Open Search & Reporting

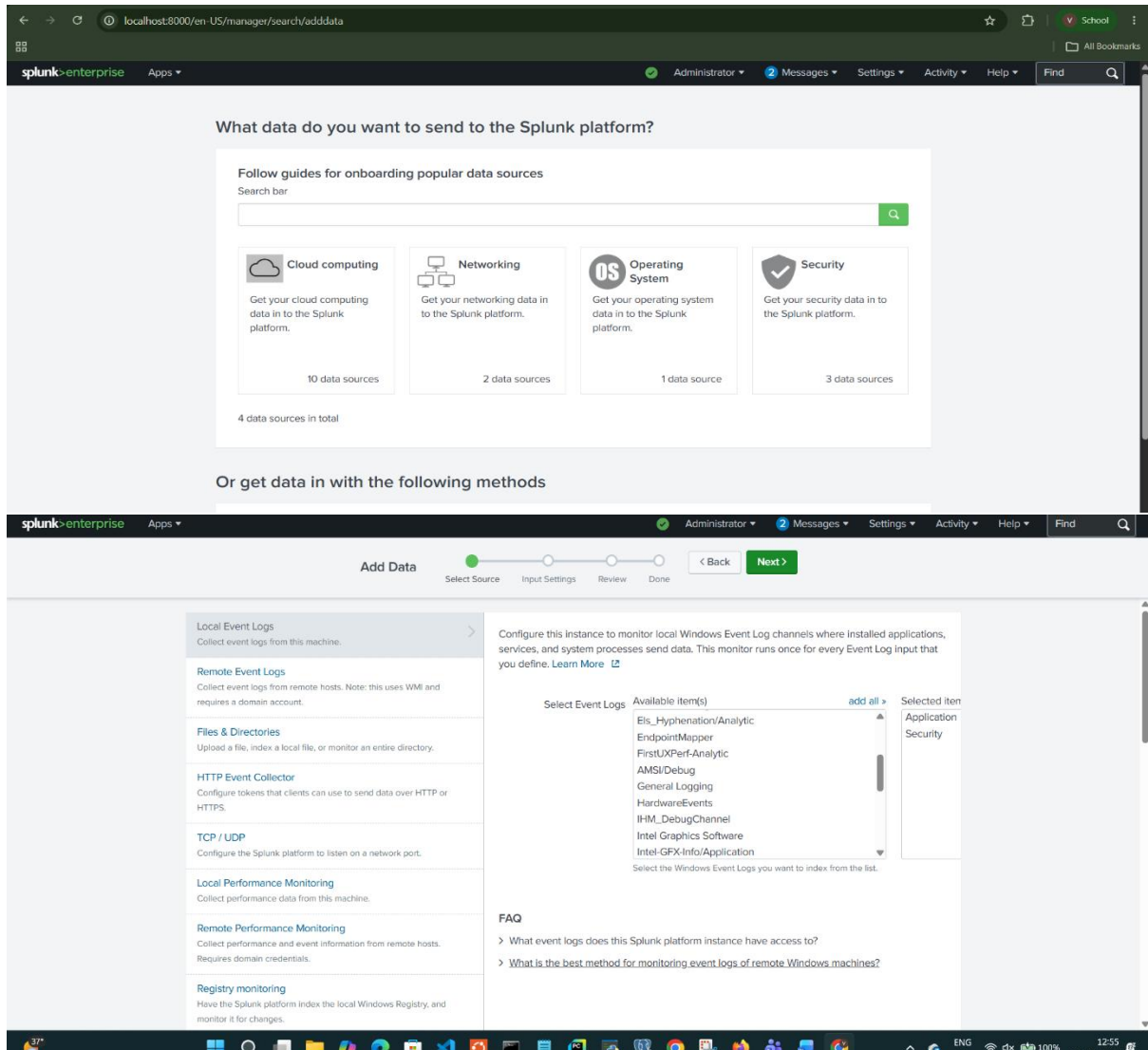
From the left navigation panel, click **Search & Reporting**.

Step 3: Use Windows Event Logs

1. Open Splunk.
2. Go to Settings → Add Data.
3. Select Monitor.

4. Choose Local Event Logs.
5. Select Windows Event Logs (Security or Application).
6. Follow the prompts to ingest the logs

Screenshot :



Step 4: Configure Input Settings

After selecting the log file:

- Review the detected source type.
- Select the destination index (usually **main**).
- Confirm the host information if required.
- Click **Review**.
- Click **Submit**.

Splunk will begin indexing the uploaded log file.

Screenshot :

The top screenshot shows the 'Input Settings' step of the 'Add Data' wizard. The progress bar indicates 'Select Source' is complete, 'Input Settings' is the current step, and 'Review' and 'Done' are pending. The 'Host' field is set to 'Ameex' and the 'Index' is set to 'Default'. The bottom screenshot shows the 'Review' step, where the configuration is summarized: Input Type: Windows Event Logs, Event Logs: Security Application, App Context: search, Host: Ameex, and Index: default. The 'Submit' button is visible in the bottom right of the wizard.

Step 5: Verify Successful Upload

After submission, Splunk displays a confirmation message indicating that the data has been indexed successfully.

Screenshot :

The screenshot shows the confirmation page after a successful upload. A green checkmark is displayed next to the message 'Local event logs input has been created successfully.' Below this message are four buttons: 'Start Searching', 'Add More Data', 'Download Apps', and 'Build Dashboards'. The progress bar at the top indicates that 'Select Source', 'Input Settings', and 'Review' are complete, and 'Done' is the current step.

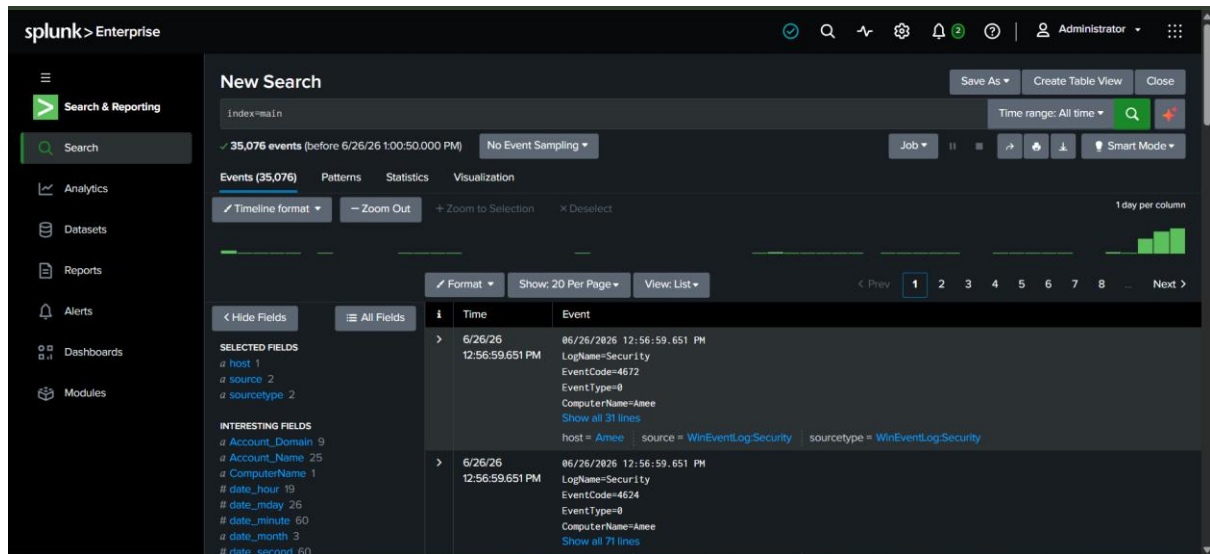
Verify Data Ingestion

To confirm that the uploaded log file has been indexed successfully, open **Search & Reporting** and execute the following query:

```
index=main
```

This query displays all events stored in the **main** index.

Screenshot :



Count Total Events

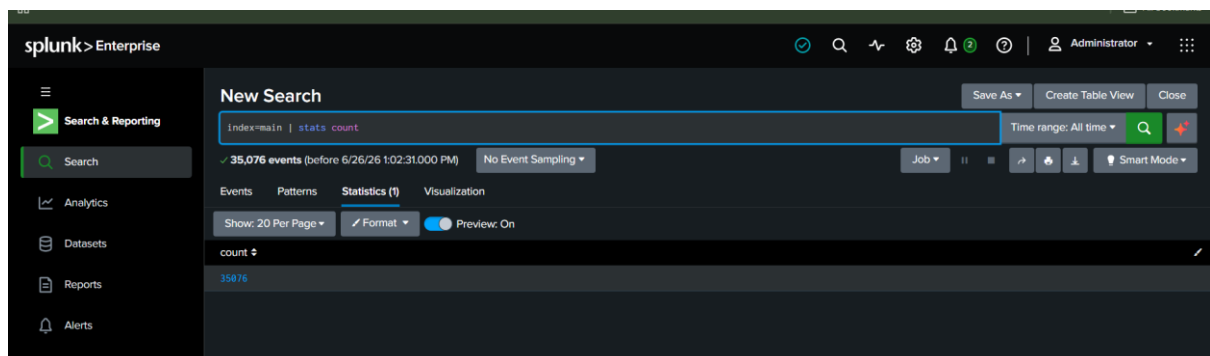
To count the total number of indexed events, execute the following SPL query:

```
index=main | stats count
```

Result: Splunk displays the total number of events available in the selected index.

Count
35076

Screenshot :



Observations

After importing the log file into Splunk, the following observations were made:

- The selected log file was uploaded successfully.
- Splunk automatically indexed the log data.
- All events became searchable using SPL queries.
- The index=main query confirmed successful data ingestion.
- The stats count command displayed the total number of events stored in the index.
- The ingested data is now ready for further analysis, dashboard creation, and alert generation.

Conclusion

The data ingestion process was successfully completed by importing a sample log file into Splunk Enterprise. The uploaded data was indexed in the **main** index and verified using SPL search queries. Splunk successfully displayed all indexed events and calculated the total event count. This confirms that the platform is ready for performing security analysis, creating dashboards, and generating alerts based on the ingested log data.

PART 4: SEARCH & ANALYSIS

Perform the following searches:

1. View all events.
2. Find failed login events.
3. Identify top source IP addresses.
4. Count unique IP addresses.
5. Find events with HTTP 404 errors.

Document your observations.

Introduction

After successfully ingesting the Windows Event Logs into Splunk Enterprise, several search queries were executed using the Splunk Processing Language (SPL). These queries were used to analyze system events, authentication activities, event distribution, and overall log statistics. The analysis helps identify security-related events and provides insights into system activity.

4.1 View All Events

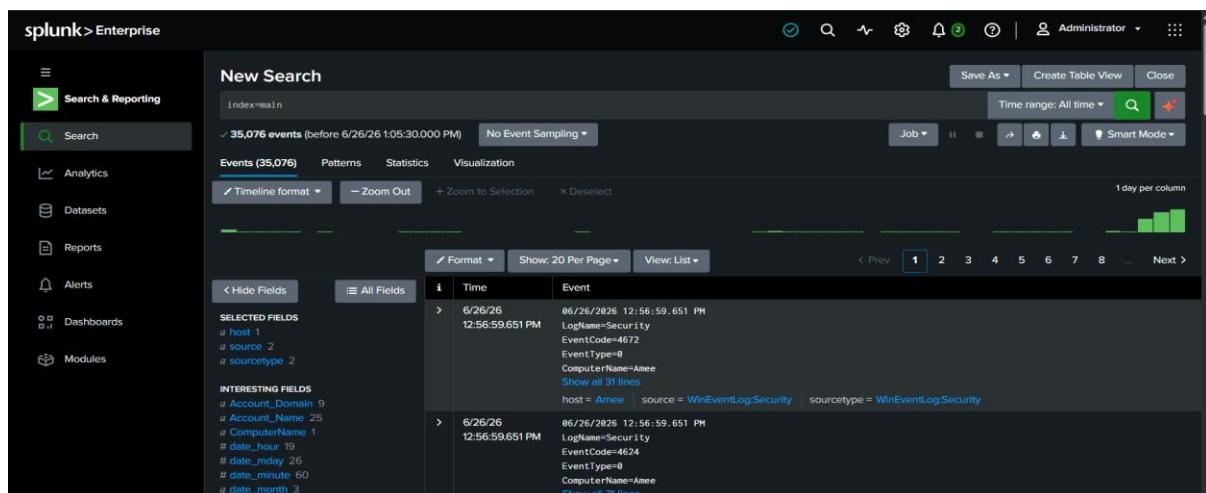
Search Query: `index=main`

Observation

This query displays all the events stored in the **main** index. The results include Application and Security event logs with details such as Event ID, timestamp, computer name, and event description.

Result: Approximately **35,076 events** were displayed.

Screenshot:



4.2 Find Failed Login Events

Search Query

index=main source="WinEventLog:Security" EventCode=4625

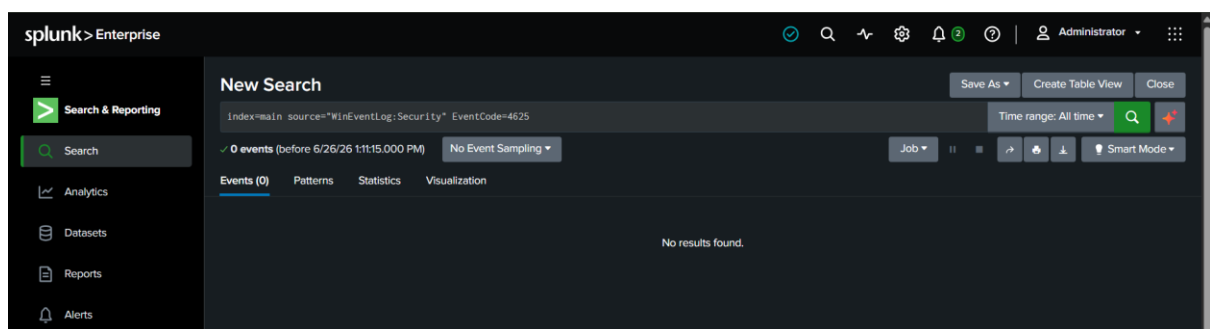
Observation

Event ID **4625** represents failed login attempts in Windows Security logs.

Result: No failed login events were found in the available log data.

This indicates that no unsuccessful authentication attempts were recorded during the selected time period.

Screenshot:



4.3 Identify Successful Login Events

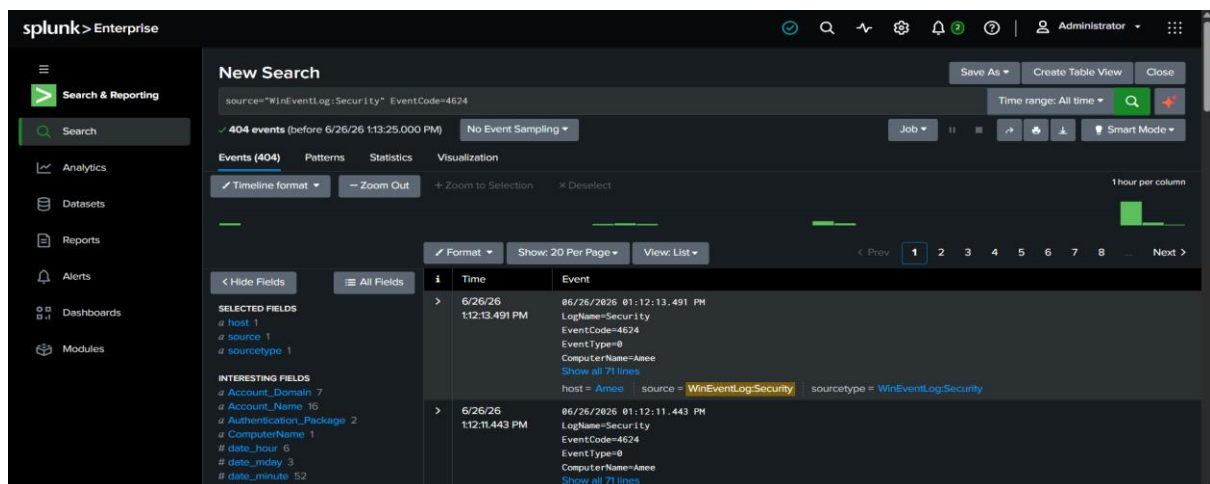
Search Query: source="WinEventLog:Security" EventCode=4624

Observation

Event ID **4624** represents successful user logins.

Several successful authentication events were observed, confirming that user login activities were being recorded correctly.

Screenshot:



4.4 Count Total Events

Search Query

index=main | stats count

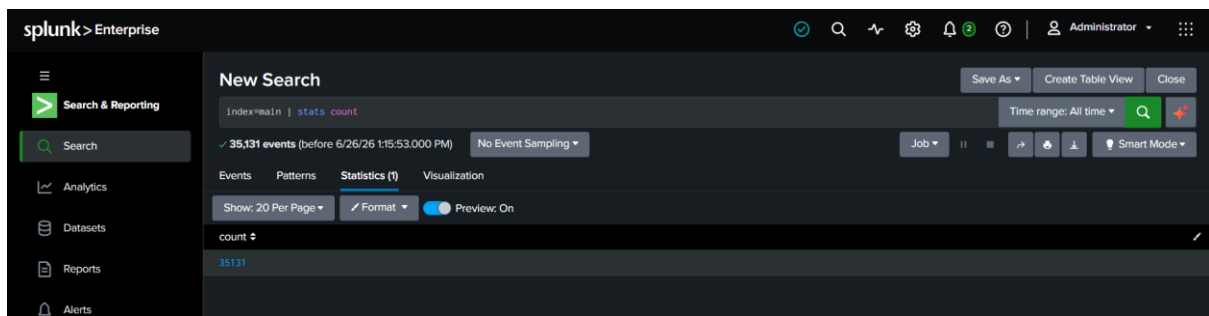
Observation

The query counts the total number of indexed events.

Result:

Count
35,131

Screenshot:



4.5 Count Unique Hosts

Since Windows Event Logs generally contain the **host** field instead of source IP addresses, the following query was used.

Search Query: index=main | stats dc(host)

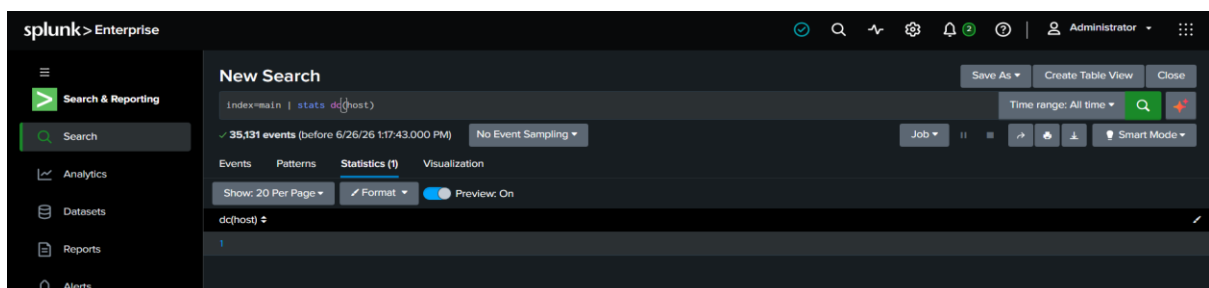
Observation

This query counts the number of unique systems sending logs to Splunk.

Result: 1

because logs were collected only from the local computer.

Screenshot:



4.6 Top Event Codes

Instead of Top Source IP Addresses, Windows logs are analyzed using Event IDs.

Search Query

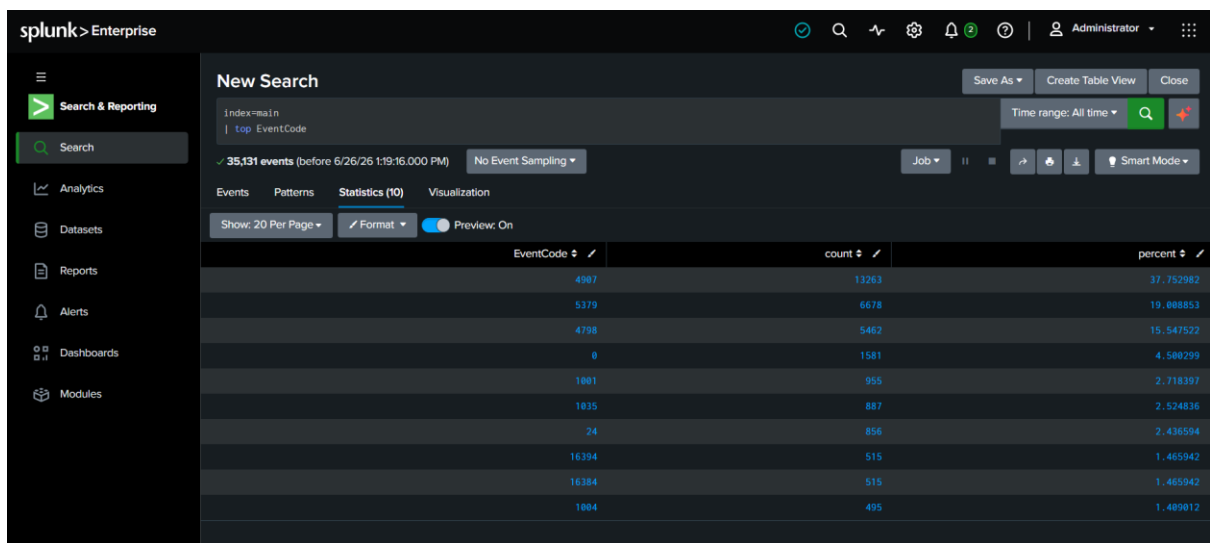
index=main

| top EventCode

Observation

The query displays the most frequently occurring Windows Event IDs.

Screenshot:



The screenshot shows the Splunk Enterprise interface with a search query 'index=main | top EventCode'. The search results are displayed in a table with columns for EventCode, count, and percent. The results show the top 10 event codes and their frequencies.

EventCode	count	percent
4987	13263	37.752982
5379	6678	19.888851
4798	5462	15.947522
0	1581	4.580299
1001	955	2.718397
1035	887	2.524836
24	856	2.436594
16394	515	1.465942
16384	515	1.465942
1004	495	1.409812

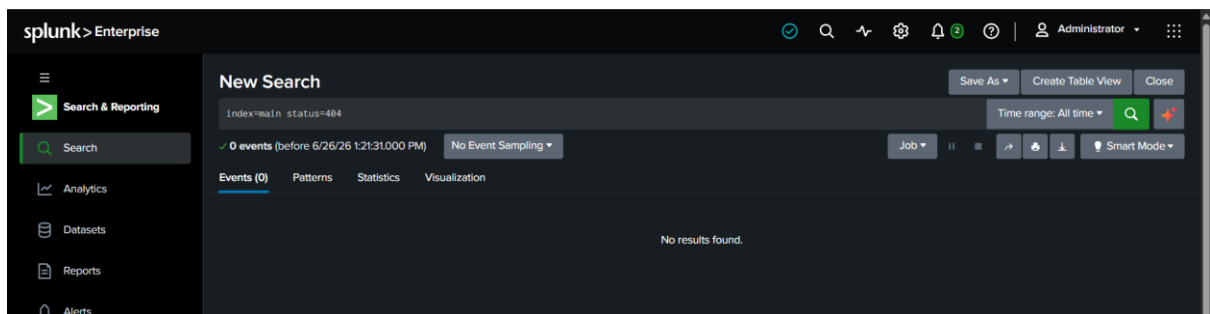
4.7 HTTP 404 Errors

Search Query: index=main status=404

Observation

No HTTP 404 events were found because Windows Security and Application logs do not record web server response codes.

Screenshot:



The screenshot shows the Splunk Enterprise interface with a search query 'index=main status=404'. The search results are displayed in a table with columns for EventCode, count, and percent. The results show 0 events found.

EventCode	count	percent
No results found.		

Findings and Observations

The following observations were made during the search and analysis process:

1. Windows Event Logs were successfully indexed into Splunk Enterprise.
2. Approximately **35,076** events were available for analysis.
3. Successful login events (Event ID **4624**) were identified in the Security logs.
4. No failed login attempts (Event ID **4625**) were detected during the selected time range.
5. One unique host was identified, indicating that the logs originated from the local system.
6. Frequently occurring Event IDs provided insight into authentication and system activities.
7. HTTP 404 errors were not available because Windows Event Logs do not contain web server status codes.

PART 5: DASHBOARD CREATION

Create a dashboard showing:

- Total Events
- Top IP Addresses
- Top Error Codes
- Event Distribution

Take screenshots.

Objective

The objective of this practical is to create a Splunk dashboard that visually represents the ingested Windows Event Logs. The dashboard provides a centralized view of important log information and helps monitor system activities through charts, tables, and statistical panels.

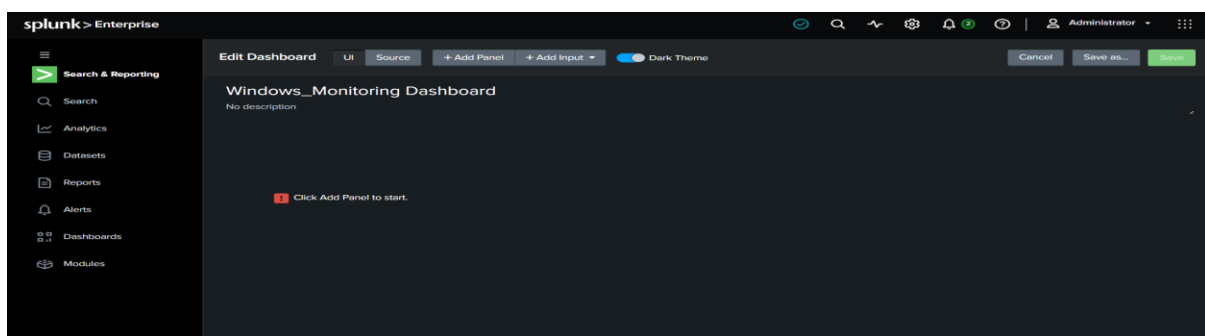
Create a New Dashboard:

1. Open **Splunk Enterprise**.
2. Click **Dashboards** from the left navigation pane.
3. Click **Create New Dashboard**.
4. Enter the following details:

Field	Value
Dashboard Title	Windows Monitoring Dashboard
Dashboard ID	windows_monitoring_dashboard
Permissions	Private
Dashboard type	Classic

5. Click **Create Dashboard**.

Screenshot :



Panel 1: Add Panel – Total Events

Procedure

1. Click **Add Panel**.
2. Select **New Panel** → Single value
3. Enter:

Panel Title

Total Events

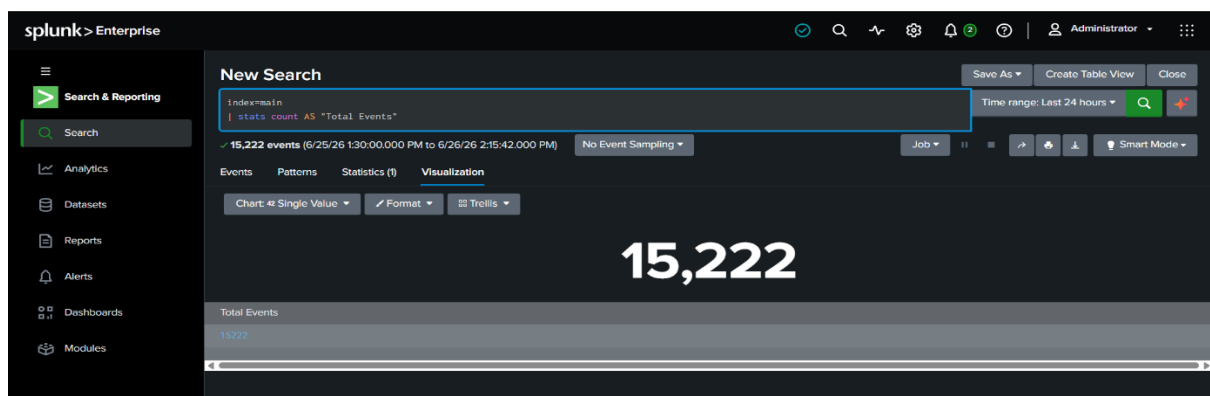
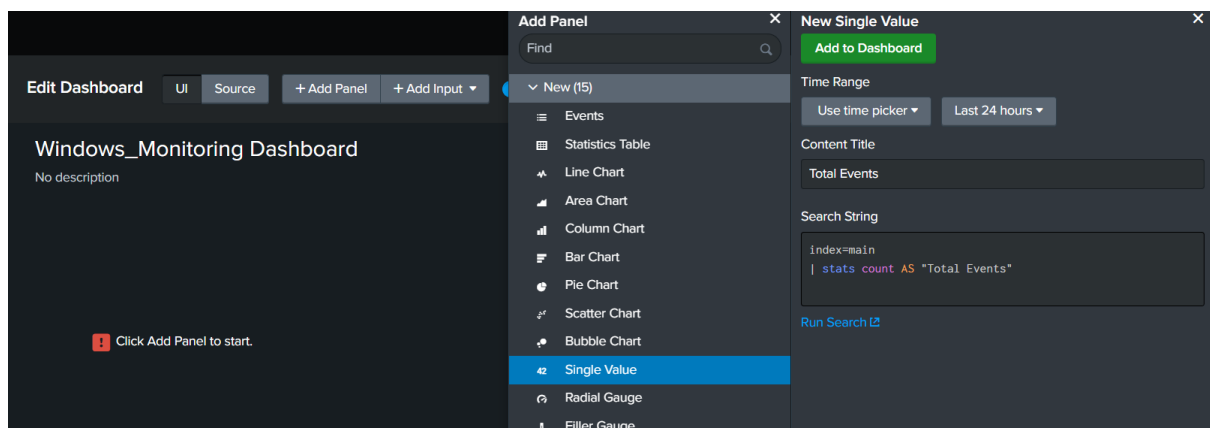
Search Query

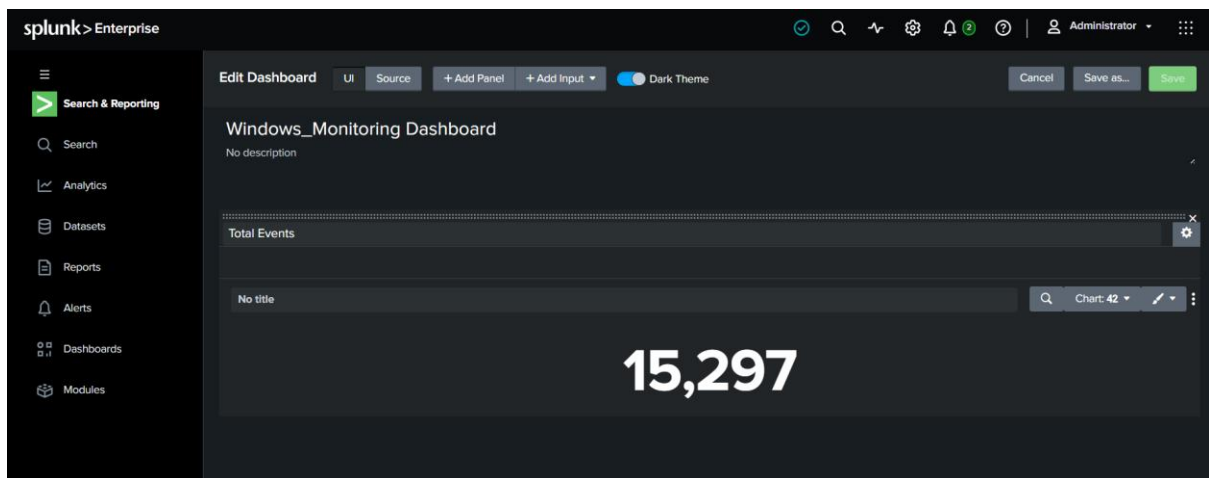
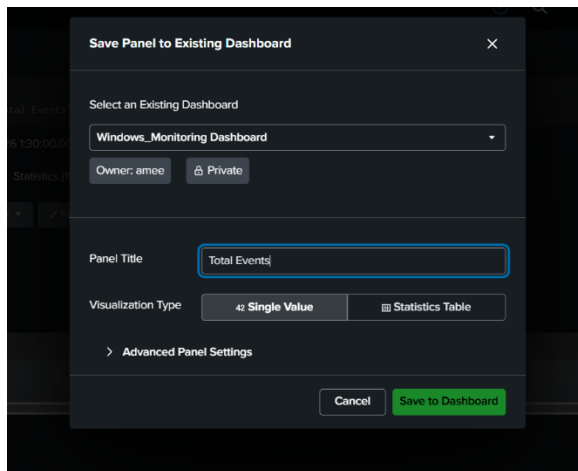
index=main

| stats count AS "Total Events"

4. Click **Run Search**.
5. Click **Visualization**.
6. Select **Single Value**.
7. Click **Save**.

Screenshot :





Panel 2: Add Panel – Top Hosts

Procedure

1. Click **Add Panel**.
2. Select **New Panel**.

Panel Title:Top Hosts

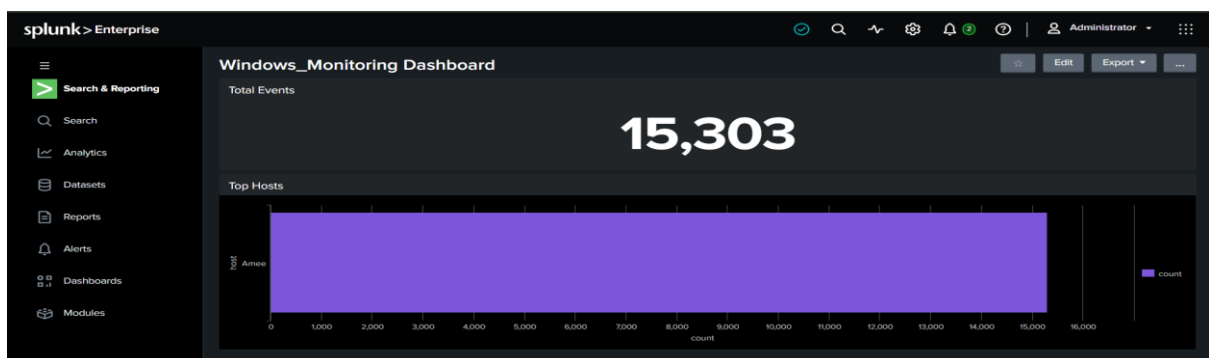
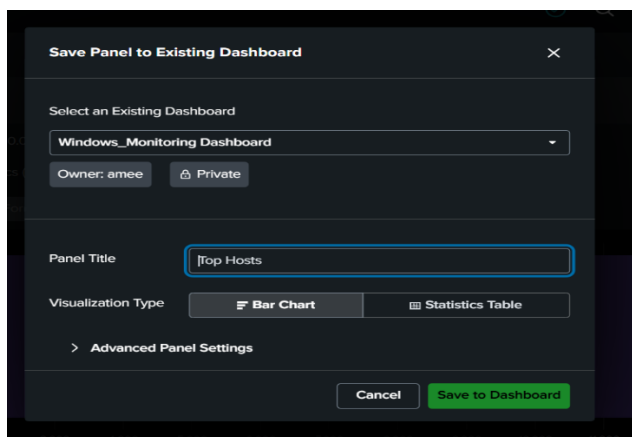
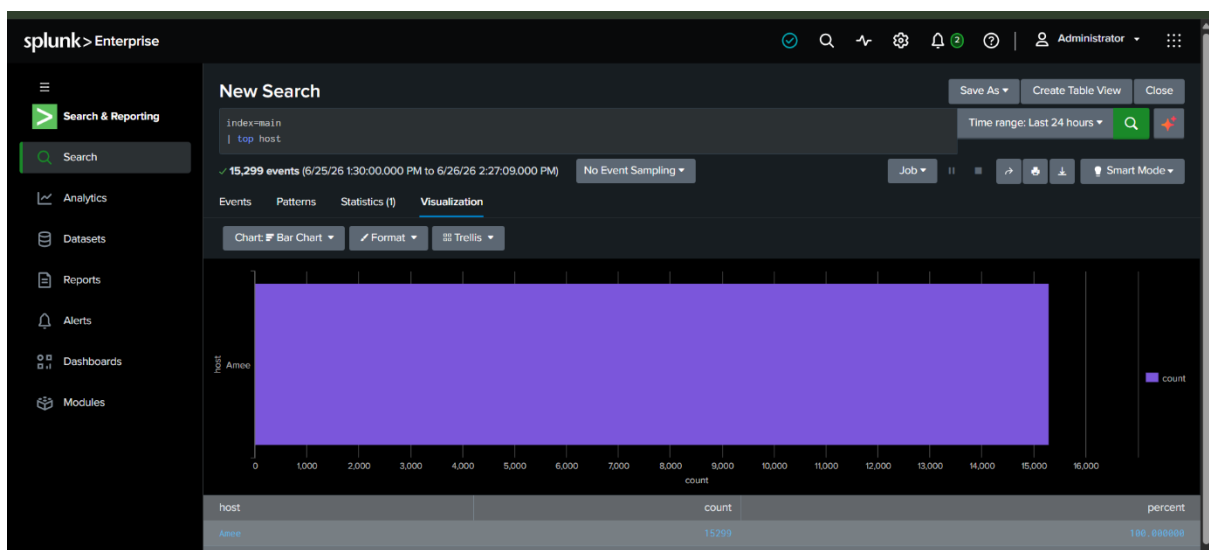
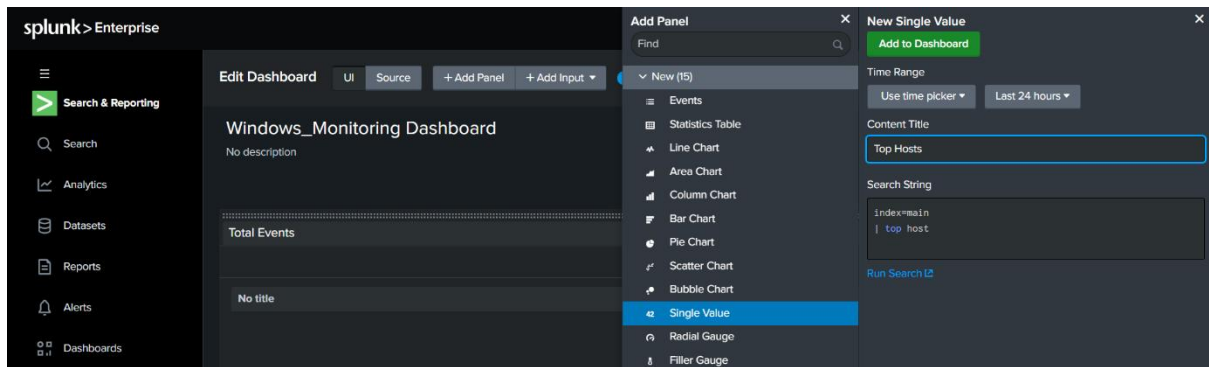
Search Query

index=main

| top host

3. Run the search.
4. Click **Visualization**.
5. Select **Bar Chart**.
6. Save the panel.

Screenshot :



Panel 3: Top IP Addresses

Objective

To identify the IP addresses that generated the highest number of events in the indexed log data.

- Open **Dashboards** and select your dashboard:

Windows Security Monitoring Dashboard

- Click **Add Panel** → **New** → **single value**

- Enter the panel details.

Panel Title

Top IP Address

- Enter the search query.

index=main

| top Source_Network_Address

- Click **Run Search**.

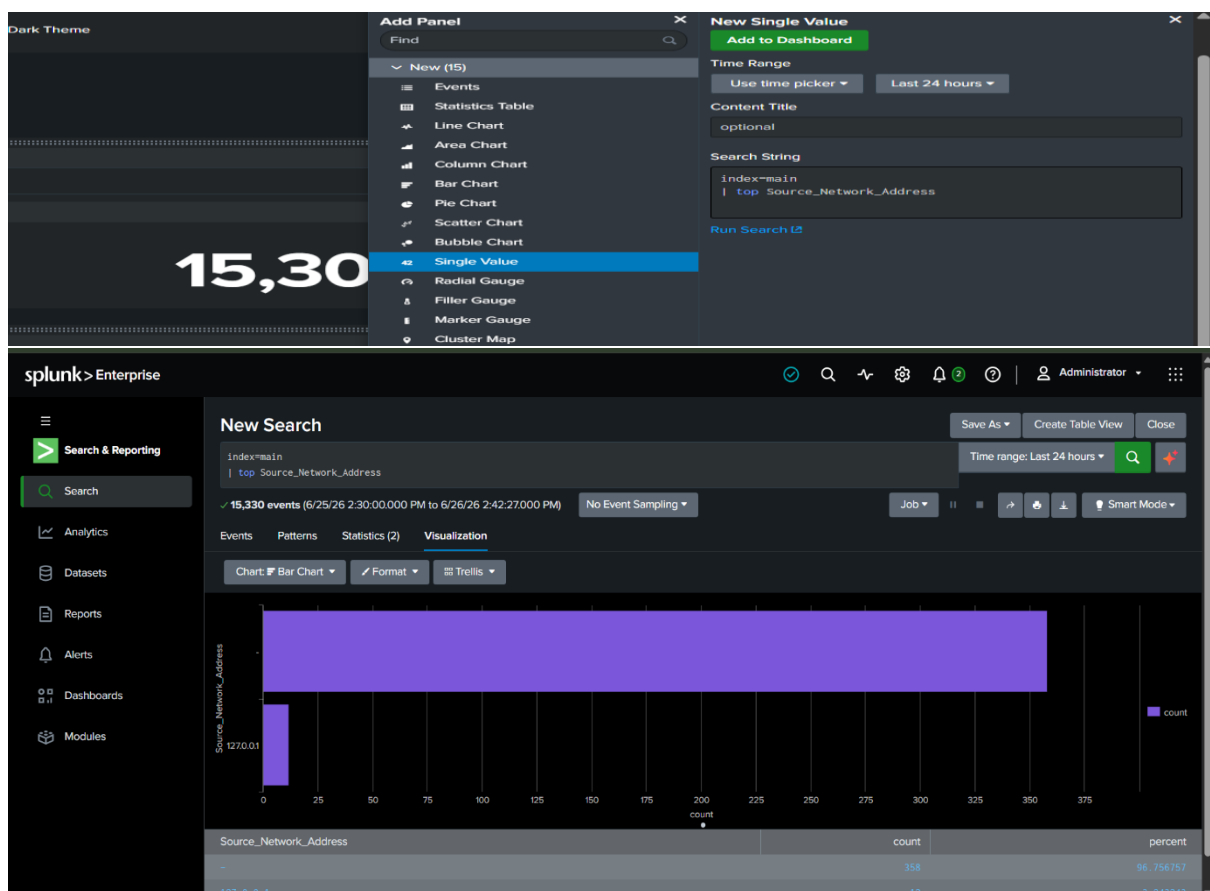
- Select the visualization.

- **Visualization:** Bar Chart

- Click **Apply**.

- Click **Save**.

Screenshot :



Save Panel to Existing Dashboard

×

Select an Existing Dashboard

Windows_Monitoring Dashboard

Owner: amee Private

Panel Title

Top IP Address

Visualization Type

Bar Chart

Statistics Table

> Advanced Panel Settings

Cancel

Save to Dashboard



Panel 4: Top Error Codes

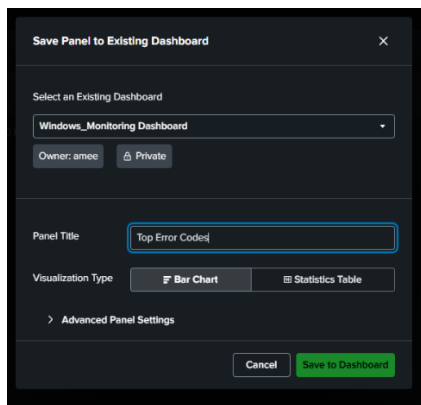
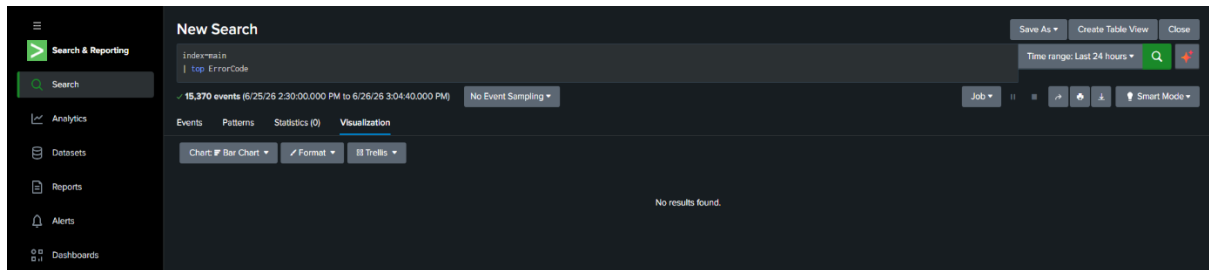
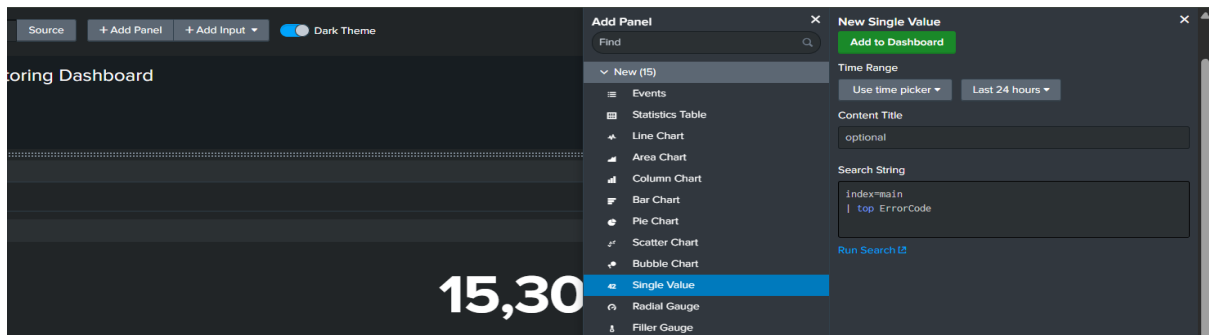
Objective

To identify the most frequently occurring error codes recorded in the indexed logs.

Procedure

- Click **Add Panel** → **New** → **Single value**
- Enter the panel title.
Top Error Codes
- Run one of the following queries depending on your data.
index=main
| top ErrorCode
- Click **Run Search**.
- Choose the visualization.
- **Visualization:** Bar Chart
- Click **Apply**.
- Click **Save**.

Screenshot :



Panel 5: Add Panel – Event Distribution

Procedure

1. Click **Add Panel**.
2. Select **New Panel**.

Panel Title

Event Distribution Over Time

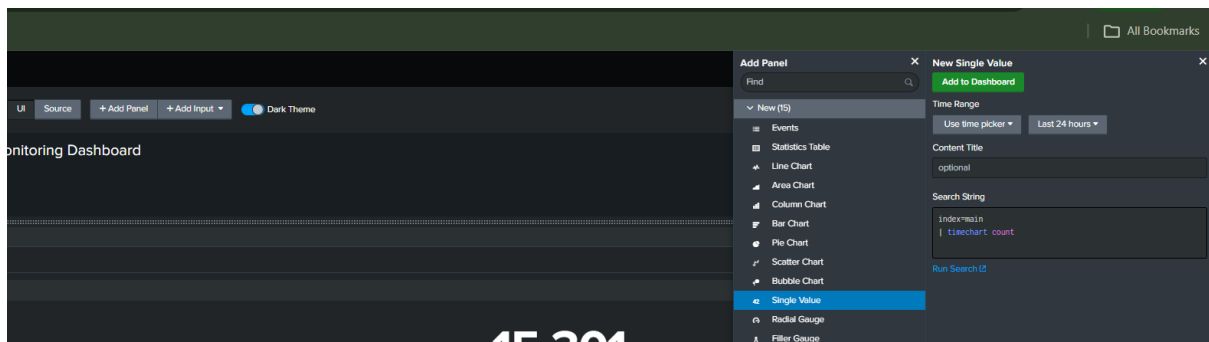
Search Query

index=main

| timechart count

3. Run the search.
4. Click **Visualization**.
5. Select **Line Chart**.
6. Save the panel.

Screenshot :



Save Panel to Existing Dashboard

Select an Existing Dashboard

Windows_Monitoring Dashboard

Owner: amee Private

Panel Title

Event Distribution Over Time

Visualization Type

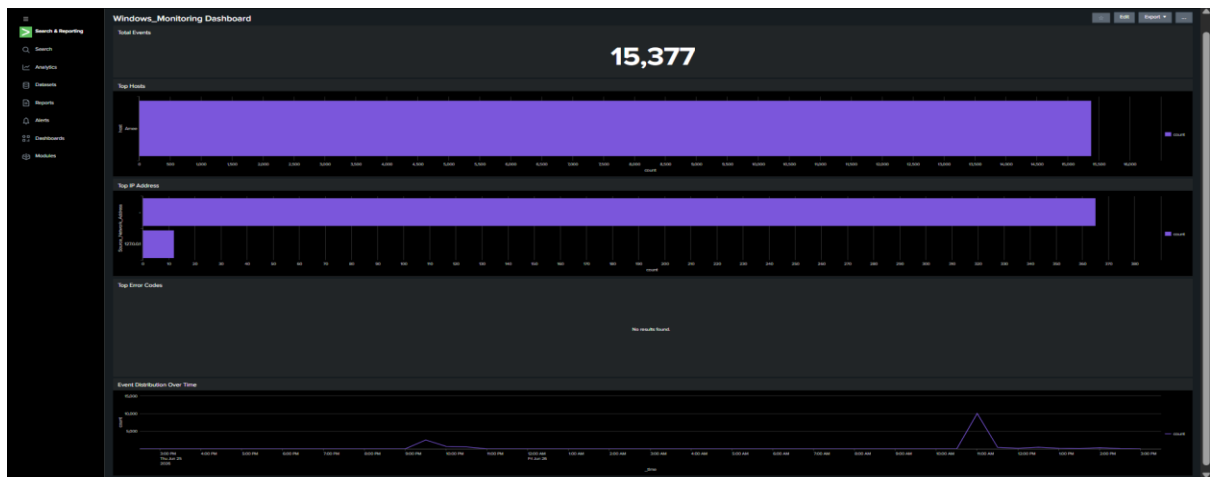
Line Chart

Statistics Table

Advanced Panel Settings

Cancel

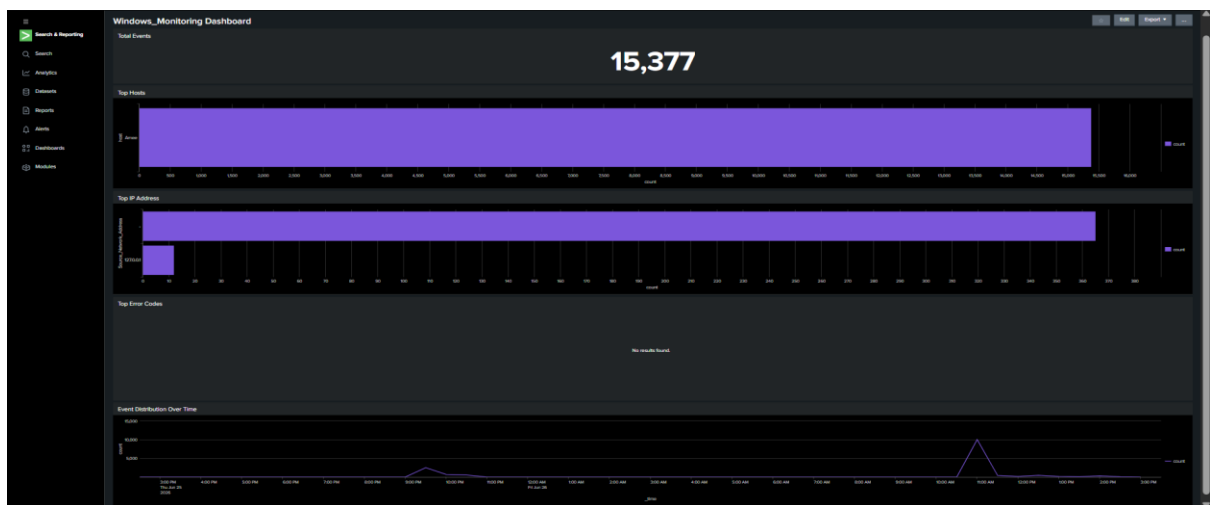
Save to Dashboard



View the Dashboard

After adding all the panels, the dashboard will display:

Screenshot :



Dashboard Summary

Panel Name	Query Used	Visualization	Purpose	Observation
Total Events	index=main stats count	Single Value	Displays the total number of indexed events.	Total events collected: 15,377 .
Top Hosts	index=main top host limit=10	Bar Chart	Shows the systems generating the highest number of logs.	Host Ame generated most of the events.
Top IP Addresses	index=main top Source_Network_Address limit=10	Bar Chart	Displays IP addresses responsible for generating events.	Most events originated from 127.0.0.1 (localhost) .
Top Error Codes	index=main top ErrorCode	Bar Chart	Displays the most frequently occurring Windows Event Codes.	No results were found because the selected query did not match any events in the indexed data.
Event Distribution Over Time	index=main timechart count	Line Chart	Shows how events are distributed over time.	Event activity remained steady with a noticeable spike around 11:00 AM .

Create an alert for:

"More than 5 failed login attempts from the same IP."

Explain:

- Why is this suspicious?
- What actions should a SOC Analyst take?

Objective

To create a Splunk alert that detects more than five failed login attempts from the same IP address. Such alerts help Security Operations Center (SOC) analysts identify possible brute-force attacks and unauthorized access attempts.

Step 1: Open Search & Reporting

1. Open **Splunk Enterprise**.
2. Navigate to **Apps** → **Search & Reporting**.
3. In the Search bar, enter the following SPL query:

```
index=main source="WinEventLog:Security" EventCode=4625
```

```
| stats count by Source_Network_Address
```

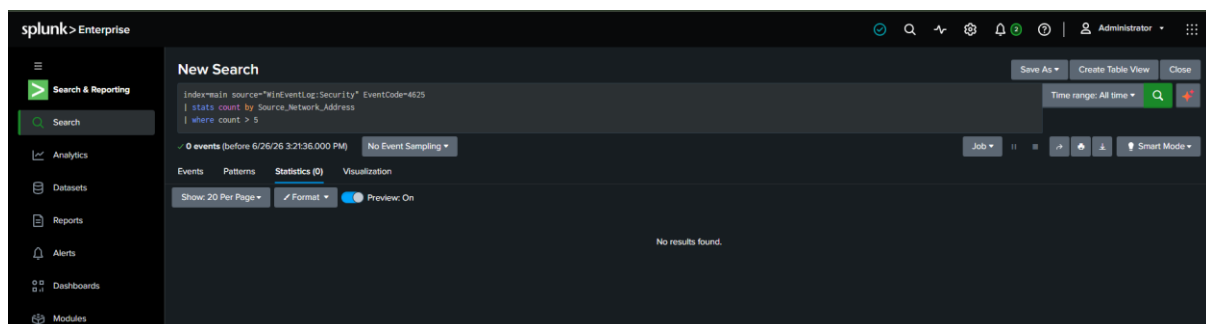
```
| where count > 5
```

4. Set the **Time Range** to **All Time**.
5. Click **Search**.

Observation

The search returned **0 events**, indicating that no failed login attempts (Event ID 4625) were present in the indexed Windows Security logs.

Screenshot :



Step 2: Save the Search as an Alert

1. Click **Save As**.
2. Select **Alert**.

Screenshot :

The screenshot shows a 'Save As Alert' dialog box with the following configuration:

- Title:** Title
- Description:** Optional
- Permissions:** Private
- Alert type:** Scheduled
- Run every week:** Run every week
- On:** Monday at 6:00
- Expires:** 24 hour(s)
- Trigger Conditions:**
 - Trigger alert when: Number of Results
 - is greater than 0
 - Trigger: Once
 - Throttle: ☐
- Trigger Actions:** + Add Actions

Buttons: Cancel, Save

Step 3: Configure the Alert

Field	Value
Title	Multiple Failed Login Attempts
Description	Detects more than five failed login attempts from the same IP address.
Permissions	Private
Alert Type	Scheduled
Run	Every 5 minutes (or Every 10 minutes if 5 minutes is unavailable)
Expires	24 hours (default)
Trigger alert when	Number of Results
Condition	is greater than 0
Trigger	Once
Throttle	Leave unchecked
Trigger Actions	Add to Triggered Alerts (leave default if no other action is required)

Screenshot :

Save As Alert

Title

Multiple Failed Login Attempts

Description

Detects more than five failed login attempts from the same IP address.

Permissions

Private

Shared in App

Alert type

Scheduled

Real-time

Run every hour

At

15

minutes past the hour

Expires

24

hour(s)

Trigger Conditions

Trigger alert when

Number of Results

Is greater than

0

Trigger

Once

For each result

Throttle

Trigger Actions

+ Add Actions

When triggered

Add to Triggered Alerts

Remove

Cancel

Save

splunk>Enterprise

Search & Reporting

Search

Analytics

Datasets

Reports

Alerts

Multiple Failed Login Attempts

Detects more than five failed login attempts from the same IP address.

Enabled: Yes. [Disable](#)

App: search

Permissions: Private. Owned by amee. [Edit](#)

Modified: Jun 26, 2026 3:32:42 PM

Alert Type: Scheduled. Hourly, at 15 minutes past the hour. [Edit](#)

Trigger Condition: Number of Results is > 0. [Edit](#)

Actions: 1 Action [Edit](#)

Add to Triggered Alerts

There are no fired events for this alert.

Why is this suspicious?

More than five failed login attempts from the same IP address within a short period may indicate a **brute-force attack**, where an attacker repeatedly tries different passwords to gain unauthorized access. It may also suggest **credential stuffing**, in which stolen usernames and passwords are used to attempt logins across multiple accounts. Repeated authentication failures from a single source are considered suspicious because they can be an early sign of unauthorized access attempts and may compromise system security if not detected promptly.

What actions should a SOC Analyst take?

A SOC Analyst should perform the following actions:

1. **Verify the Alert:** Confirm that the failed login attempts are genuine and not caused by user error or system misconfiguration.
2. **Identify the Source IP:** Determine the IP address responsible for the repeated failed login attempts and verify whether it belongs to an internal or external network.
3. **Analyze Related Logs:** Review Windows Security logs, firewall logs, VPN logs, and authentication records to gather additional evidence.
4. **Check User Accounts:** Identify the targeted user accounts and determine whether any successful login occurred after the failed attempts.
5. **Contain the Threat:** Block or temporarily restrict the suspicious IP address if malicious activity is confirmed.
6. **Secure the Account:** Advise affected users to reset their passwords and enable Multi-Factor Authentication (MFA) if available.
7. **Escalate the Incident:** Report the incident to the incident response team and follow the organization's security procedures.
8. **Continue Monitoring:** Monitor the environment for additional suspicious login attempts or related malicious activities to ensure the threat has been contained.